

Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{z}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and – where linear – the tilt angle ψ measured from $\hat{x}$.

(a) $\vec{E} = \hat{x}3\cos(\omega t - kz)$

(b) $\vec{E} = \hat{x}2\cos(\omega t - kz) + \hat{y}2\cos(\omega t - kz)$

(c) $\vec{E} = \hat{x}5\cos(\omega t - kz) + \hat{y}5\cos(\omega t - kz - 90^\circ)$

(d) $\vec{E} = \hat{x}5\cos(\omega t - kz) + \hat{y}5\cos(\omega t - kz + 90^\circ)$

(e) $\vec{E} = \hat{x}3\cos(\omega t - kz) + \hat{y}1\cos(\omega t - kz - 90^\circ)$

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with $AR \leq 3$ dB across its operating band.

(a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

AR =

(b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{x}$.

(c) An $\hat{x}$ receiver picks off E_x ; a $\hat{y}$ receiver picks off E_y . Compute the ratio of received power in dB. Should it match part (a)?

$P_x/P_y =$

3. Polarization loss factor.

- (a) A ground station transmits a *linearly* polarized signal along $\hat{z}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.

PLF =

- (b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.

PLF =

- (c) Two linear whip antennas: radio A vertical, radio B tilted $\psi = 30^\circ$ from vertical. Compute the PLF.

PLF =

- (d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd "VSWR ≤ 2 from 2.30GHz to 2.55GHz."

(a) What is the impedance bandwidth in MHz?

BW =

(b) What is the fractional bandwidth (FBW) in percent?

FBW =

(c) Convert VSWR = 2 to $|\Gamma|$ and to return loss in dB.

$|\Gamma|$ =

RL =

(d) Narrowband, broadband, or ultra-wideband? Name the likely antenna family, using the bandwidth table from Lesson 3.

5. Fractional and ratio bandwidth.

- (a) A cellular antenna covers 698–960MHz (low) and 1.71–2.69GHz (high). Compute the FBW and RBW of each band.

low =

high =

- (b) A UWB radar antenna covers 3.1–10.6GHz. Compute its RBW. Does it meet the FCC UWB threshold of $\text{RBW} \geq 2:1$?

RBW =

- (c) Which antenna family would you pick for each of the three bands? Give a one-sentence reason for each that names the *mechanism* buying the bandwidth – resonant, traveling-wave, or self-scaling.

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915$ MHz (ISM band).

(a) Compute λ , k , and the electrical size ka .

 $\lambda =$ $k =$ $ka =$

(b) Estimate the minimum Q from $Q_{\min} \approx \frac{1}{(ka)^3} + \frac{1}{ka}$.

 $Q_{\min} =$

(c) Estimate the maximum fractional bandwidth $\text{FBW}_{\max} \approx 1/Q_{\min}$ (for $\text{VSWR} \leq 2$).

 $\text{FBW}_{\max} =$

(d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

7. **Reading the datasheet.** A vendor offers a GPS patch specified as **RHCP**, $AR \leq 3 \text{ dB}$, “bandwidth 4% (VSWR ≤ 2)”. You are choosing the antenna for the other end of the link.

(a) The datasheet defines handedness *from the receiver looking back toward the incoming wave*. Which convention is that, what is the part’s sense in **IEEE** terms, and what goes wrong if you pair it with a satellite transmitting IEEE RHCP?

(b) Convert $AR = 3 \text{ dB}$ to a linear ratio A , and give A^2 .

$A =$

(c) A *linear* antenna receives from this patch. Using

$$PLF(\psi) = \frac{A^2 \cos^2 \psi + \sin^2 \psi}{A^2 + 1}, \text{ find the best and worst PLF over all orientations } \psi, \text{ in dB.}$$

best =

worst =

(d) Show that the peak-to-null swing in dB equals the axial ratio in dB.

- (e) The far end is an opposite-sense CP antenna, also $AR = 3 \text{ dB}$, major axes aligned. Worst-case rejection is $\left(\frac{A^2 - 1}{A^2 + 1}\right)^2$. Evaluate it and compare with the idealized PLF table.

rejection =

- (f) Name two things you must ask before believing the antenna works over the advertised 4%.

Documentation: _____